# Assignment 1

Week 1

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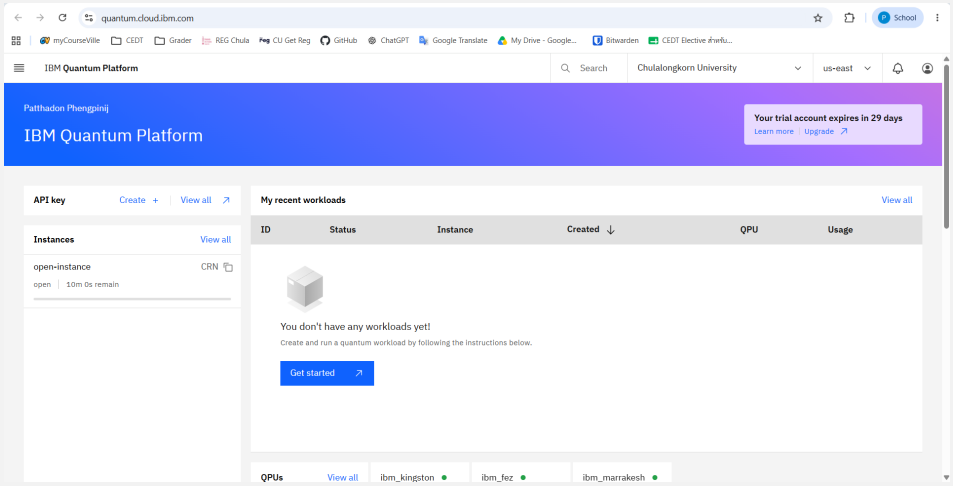
### Problem 1. Quantum Computing Account

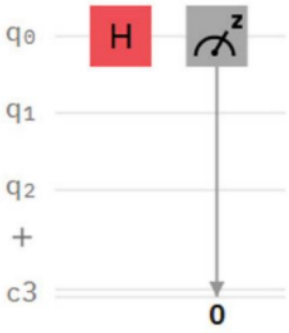
Please go to IBM Quantum Experience and register an account, https://quantum.cloud.ibm.com/.

**Solution.** Logged in to IBM Quantum Experience and registered an account successfully.



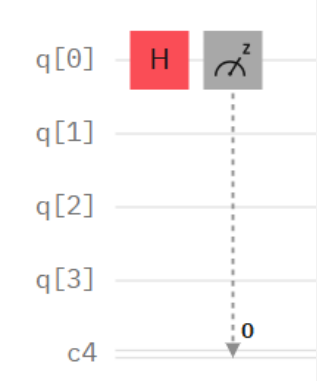

### Problem 2. My First Quantum Circuit

Try to create a circuit. Click on IBM Quantum Composer and drag the gates to form the following circuit.



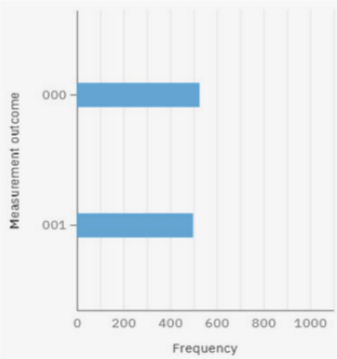

The first icon is a quantum gate called Hadamard gate and the second icon is a measurement. The input is from the left ($q_0$). Do not worry if you do not understand what it is at this moment. But good job! You have created your first quantum circuit!

**Solution.** Finished creating the quantum circuit as instructed.



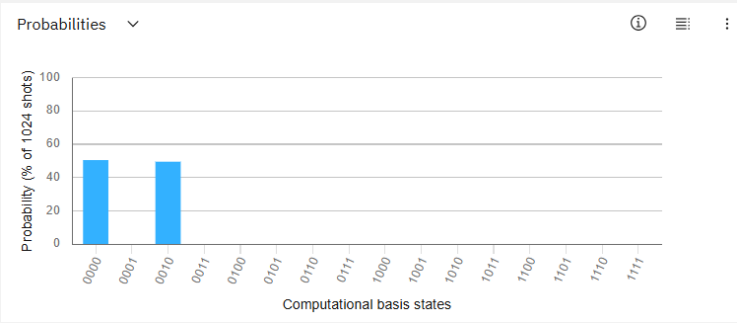

### Problem 3. My First Quantum Computing

Submit the job. Click on the result when it is finished. You see some like this. You may submit to a simulator or real quantum hardware.




We started with a state in $|0\rangle$ (a basis state). After the *Hadamard gate*, it becomes a superposition of $|0\rangle$ and $|1\rangle$, i.e. $\sim |0\rangle + |1\rangle$ (we will learn later). Then we performed a measurement on the output and the superposition state collapses to either $|0\rangle$ or $|1\rangle$. The histogram shows that about 50% of the chance we obtain $|0\rangle$ and another 50% of the chance we obtain $|1\rangle$. Cool! We have done our first quantum computation. This job was submitted to a real quantum computer, called "ibmq_casablanca." It is a 5-qubit quantum computer. However, this computer is gone. But definitely, you will find other more powerful ones.

**Solution.** The job was done successfully and the result was obtained as instructed.



### Problem 4. Google Colab and Python

We need to do a lot of matrix operations. We will choose Google Colab, in which we can run Python easily. Since IBM Quantum Experience also uses Python, it will save your efforts. Please open a Google Colab account here https://colab.research.google.com/.

**Solution.** I decided to use my local computer to run the Jupyter notebook instead of Google Colab.

### Problem 5. Vector Inner Product Using Python

We use the NumPy library to perform vector and matrix operations. To import the library, type

```
import numpy as np
```

Then let's implement

$$\vec{v_1} \cdot \vec{v_2} = \begin{pmatrix} \sqrt{3} & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \left(\sqrt{3} \times 1\right) + (1 \times 1) = \sqrt{3} + 1 \approx 2.732$$

```
v1 = np.array([np.sqrt(3), 1])
v2 = np.array([1, 1])
inner_product = np.vdot(v1, v2)

print(inner_product)
```

Type Shift-Enter to let it run. Do you see the same result? Note that we use `np.vdot` instead of `np.dot` because we need to use complex vector dot products in quantum computing. Try also `print(v1.shape)`, which will give you (2, 1). It means we constructed a $2 \times 1$ matrix, which is a column vector with 2 elements.

**Solution.** Try running the code in a Python environment and obtained the same result of approximately 2.732.

```
import numpy as np

v1 = np.array([[np.sqrt(3)], [1]])
v2 = np.array([[1], [1]])
inner_product = np.vdot(v1, v2)

print(f"Inner product: {inner_product:.3f}")
print("v1 Shape:", v1.shape)
print("v2 Shape:", v2.shape)
```

### Problem 6. Inner Product with Complex Coefficient 1

Now, let's try $\vec{v_1} = \begin{pmatrix} i \\ 1 \end{pmatrix}$ and $\vec{v_2} = \begin{pmatrix} -i \\ 1 \end{pmatrix}$. Firstly, try to use hand calculation. Then use Python to check if it is correct. Note that in the NumPy library, you need to put $i$ as $j$. For example, $5+i$ should be entered as $5 + 1j$ (putting the "1" is necessary).

Think about that before reading the code below:

```python
v1 = np.array([[1j], [1]])
v2 = np.array([[-1j], [1]])
inner_result = np.vdot(v1, v2)
print(inner_result)
```

**Solution.** First, try to calculate the inner product manually:

$$\begin{aligned}
\vec{v_1} \cdot \vec{v_2} &= \begin{pmatrix} i \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -i \\ 1 \end{pmatrix} \\
&= \begin{pmatrix} i^* & 1^* \end{pmatrix} \begin{pmatrix} -i \\ 1 \end{pmatrix} \\
&= \begin{pmatrix} -i & 1 \end{pmatrix} \begin{pmatrix} -i \\ 1 \end{pmatrix} \\
&= (-i \times -i) + (1 \times 1) \\
&= -1 + 1 \\
\vec{v_1} \cdot \vec{v_2} &= 0
\end{aligned}$$

Then, re-check the result using Python code given by the problem. We can see that the result is also 0, which is the same as what we calculated manually.

**Problem 7. Inner Product with Complex Coefficient 2**

Now, let's try $\vec{v_1} = \begin{pmatrix} 1+2i \\ 1 \end{pmatrix}$ and $\vec{v_2} = \begin{pmatrix} 1-i \\ 1 \end{pmatrix}$.

**Solution.** First, we calculate the inner product manually:

$$\begin{aligned}
\vec{v_1} \cdot \vec{v_2} &= \begin{pmatrix} 1+2i \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1-i \\ 1 \end{pmatrix} \\
&= \begin{pmatrix} (1+2i)^* & 1^* \end{pmatrix} \begin{pmatrix} 1-i \\ 1 \end{pmatrix} \\
&= \begin{pmatrix} 1-2i & 1 \end{pmatrix} \begin{pmatrix} 1-i \\ 1 \end{pmatrix} \\
&= (1-2i)(1-i) + (1)(1) \\
&= 1 - i - 2i + 2i^2 + 1 \\
&= 1 - 3i - 2 + 1 \\
\vec{v_1} \cdot \vec{v_2} &= -3i
\end{aligned}$$

Then, re-check the result using the following Python code:

```python
v1 = np.array([[1 + 2j], [1]])
v2 = np.array([[1 - 1j], [1]])
inner_result = np.vdot(v1, v2)

print(f"Inner product: {inner_result:.3f}")
```

We can see that the result is also -3i, which is the same as what we calculated manually.

# APPENDIX 1

## Assignment 01 | Code Solution

# assignment-01-code

August 13, 2026

## 1 Assignment I

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### 1.1 Import External Libraries

[1]:
```
import numpy as np
```

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### 1.2 5. Vector Inner Product Using Python

[2]:
```
v1 = np.array([[np.sqrt(3)], [1]])
v2 = np.array([[1], [1]])
inner_product = np.vdot(v1, v2)

print(f"Inner product: {inner_product:.3f}")
print("v1 Shape:", v1.shape)
print("v2 Shape:", v2.shape)
```

```
Inner product: 2.732
v1 Shape: (2, 1)
v2 Shape: (2, 1)
```

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### 1.3 6. Inner Product with Complex Coefficient 1

[3]:
```
v1 = np.array([[1j], [1]])
v2 = np.array([[-1j], [1]])
inner_result = np.vdot(v1, v2)

print(f"Inner product: {inner_result:.3f}")
```

```
Inner product: 0.000+0.000j
```

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### 1.4 7. Inner Product with Complex Coefficient 2

[4]:
```
v1 = np.array([[1 + 2j], [1]])
v2 = np.array([[1 - 1j], [1]])
inner_result = np.vdot(v1, v2)

print(f"Inner product: {inner_result:.3f}")
```

```
Inner product: 0.000-3.000j
```

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